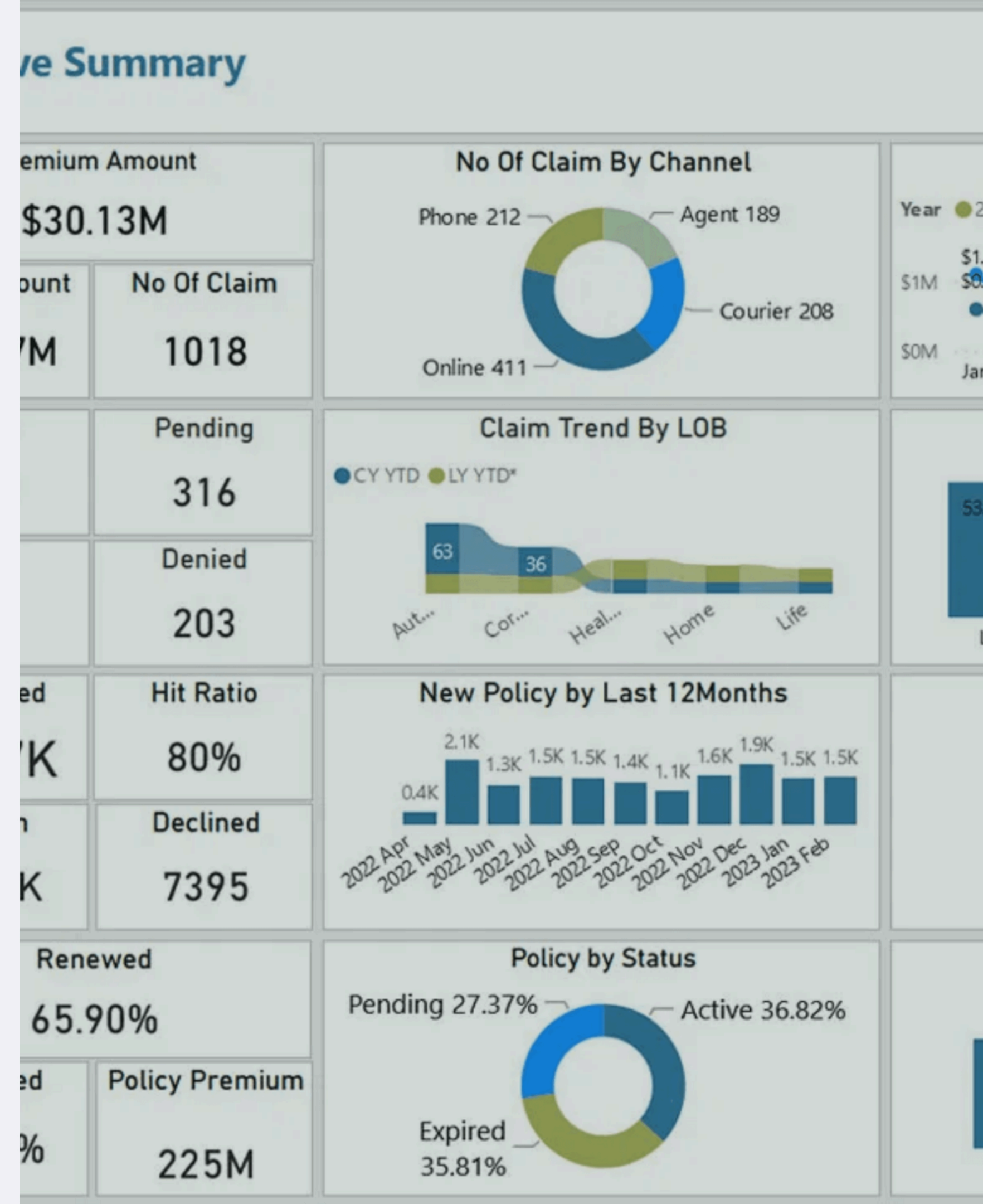


RASHAD Risk Intelligence

Intelligent Value Index (IVI)

Analytics Engine

Comprehensive Risk Assessment & Strategic Action Framework





فهرس المحتويات

01	عنوان
02	عنوان
03	عنوان
04	عنوان
05	عنوان
06	عنوان
07	عنوان
08	عنوان

Executive Summary

Objective

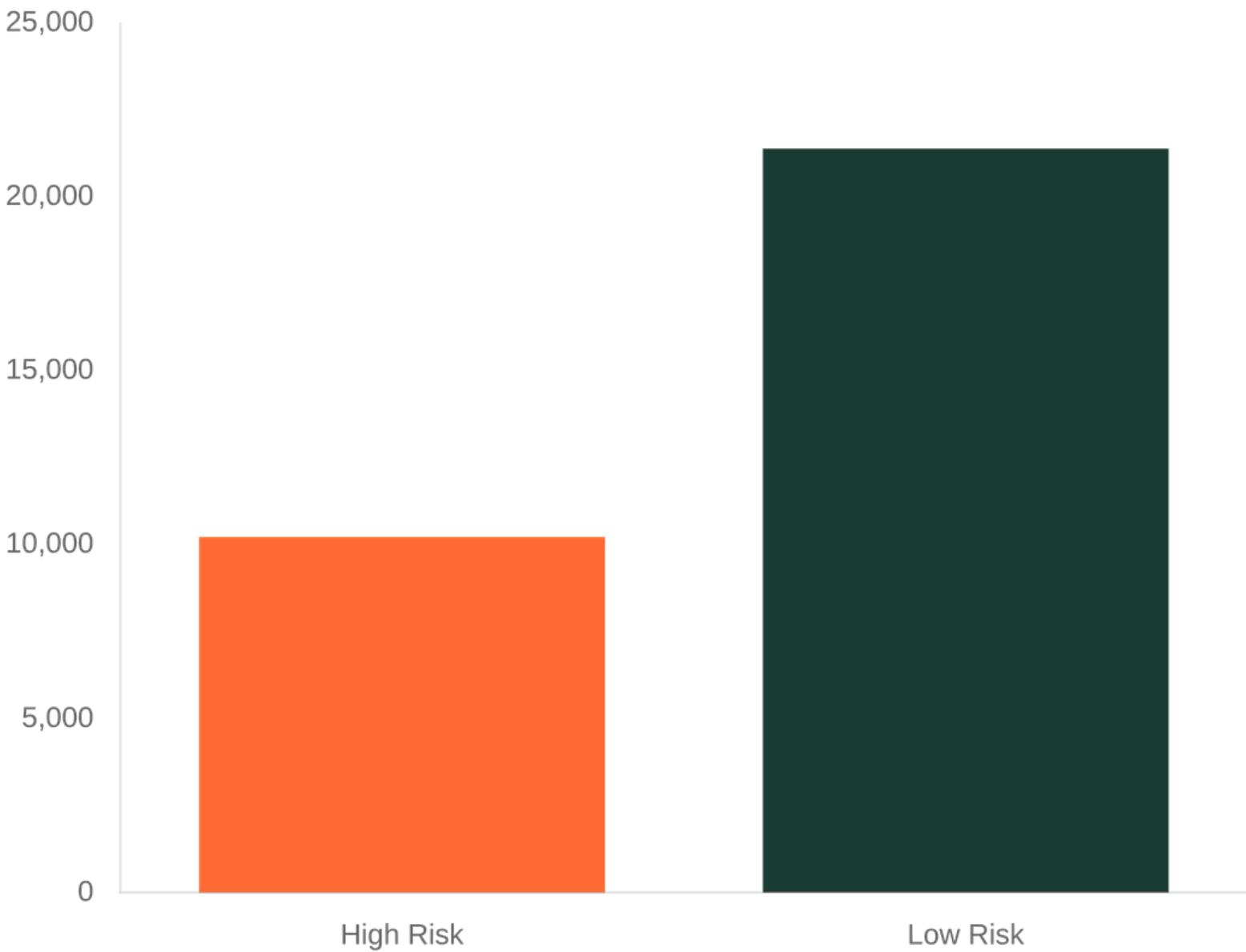
Develop a data-driven risk scoring framework for 31,617 insurance contracts.

Solution

Intelligent Value Index (IVI) combining Health, Experience, and Utilization.

Impact

Identified 32.3% high-risk contracts, enabling early intervention and cost optimization.



Portfolio Risk Segmentation

Model Logic & Methodology

$$IVI = 0.294 \times H + 0.115 \times E + 0.591 \times U$$

Health (H) - 29.4%

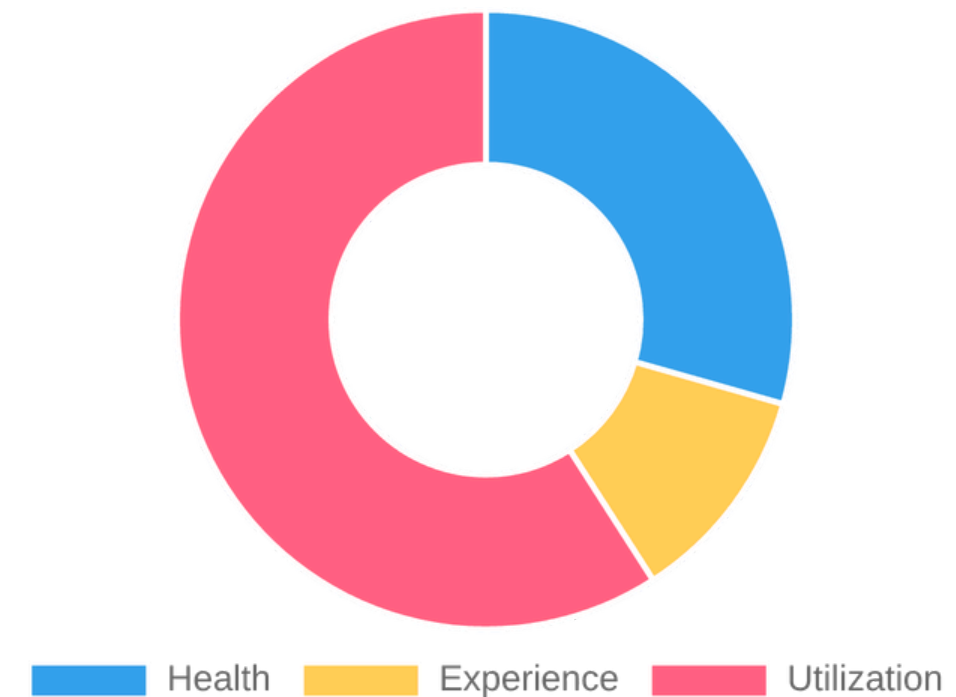
Medical complexity and clinical risk.

Experience (E) - 11.5%

Member satisfaction and service quality.

Utilization (U) - 59.1%

Cost efficiency and consumption patterns.



Data-Driven Dimension Weights

Dimension Deep Dive

Health (29.4%)

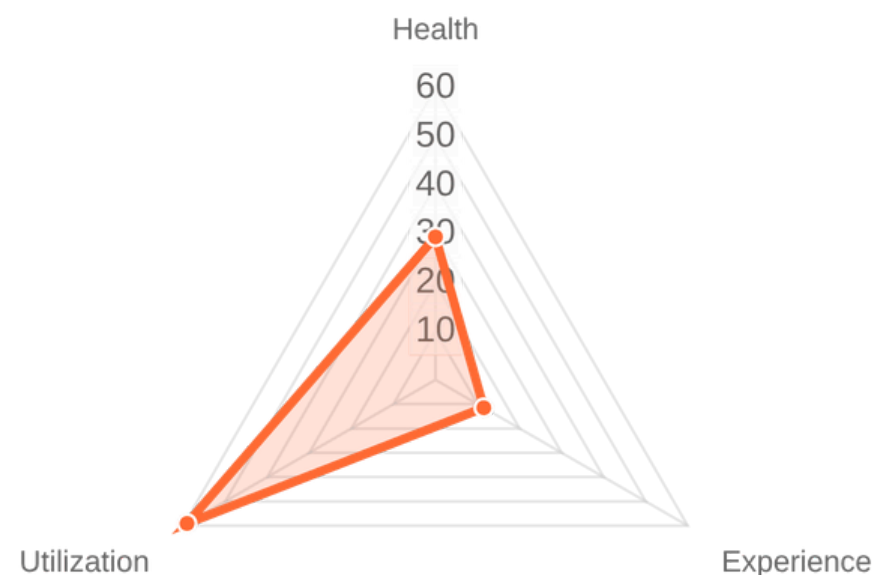
Chronic condition burden
High-risk diagnosis rate
Health risk escalation

Experience (11.5%)

Rejection rate
Complaint frequency
Service quality delays

Utilization (59.1%)

Cost per member
High-cost service rate
Services per episode



Pain Point Analysis

Utilization is the primary driver of risk, nearly twice as impactful as medical risk. Strategic focus must shift to operational efficiency.

Portfolio Risk Distribution

31,617

Total Contracts

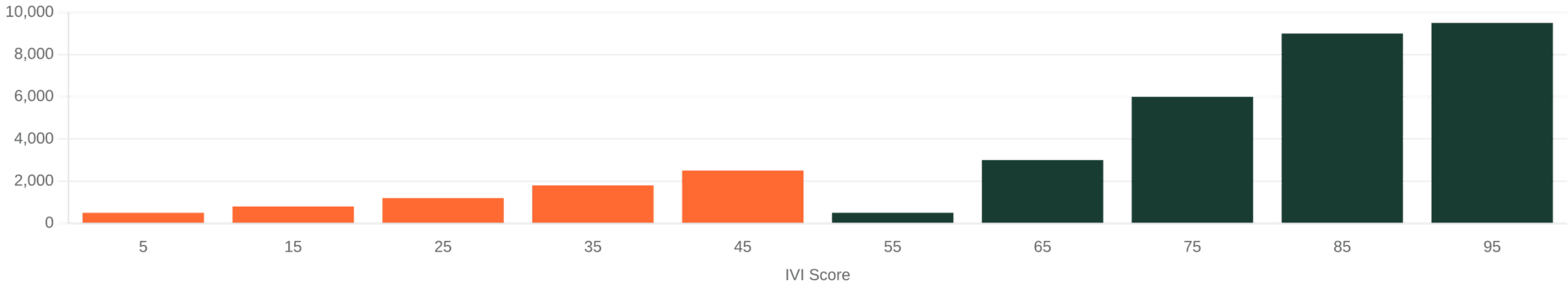
32.3%

High Risk

67.7%

Low Risk

IVI Score Distribution (Portfolio-wide)



Key Finding: The portfolio exhibits a bimodal distribution. Contracts are polarized between high-performing (75-95) and high-risk (5-50) segments, with virtually no middle ground.

Predictive Modeling

Target Variable

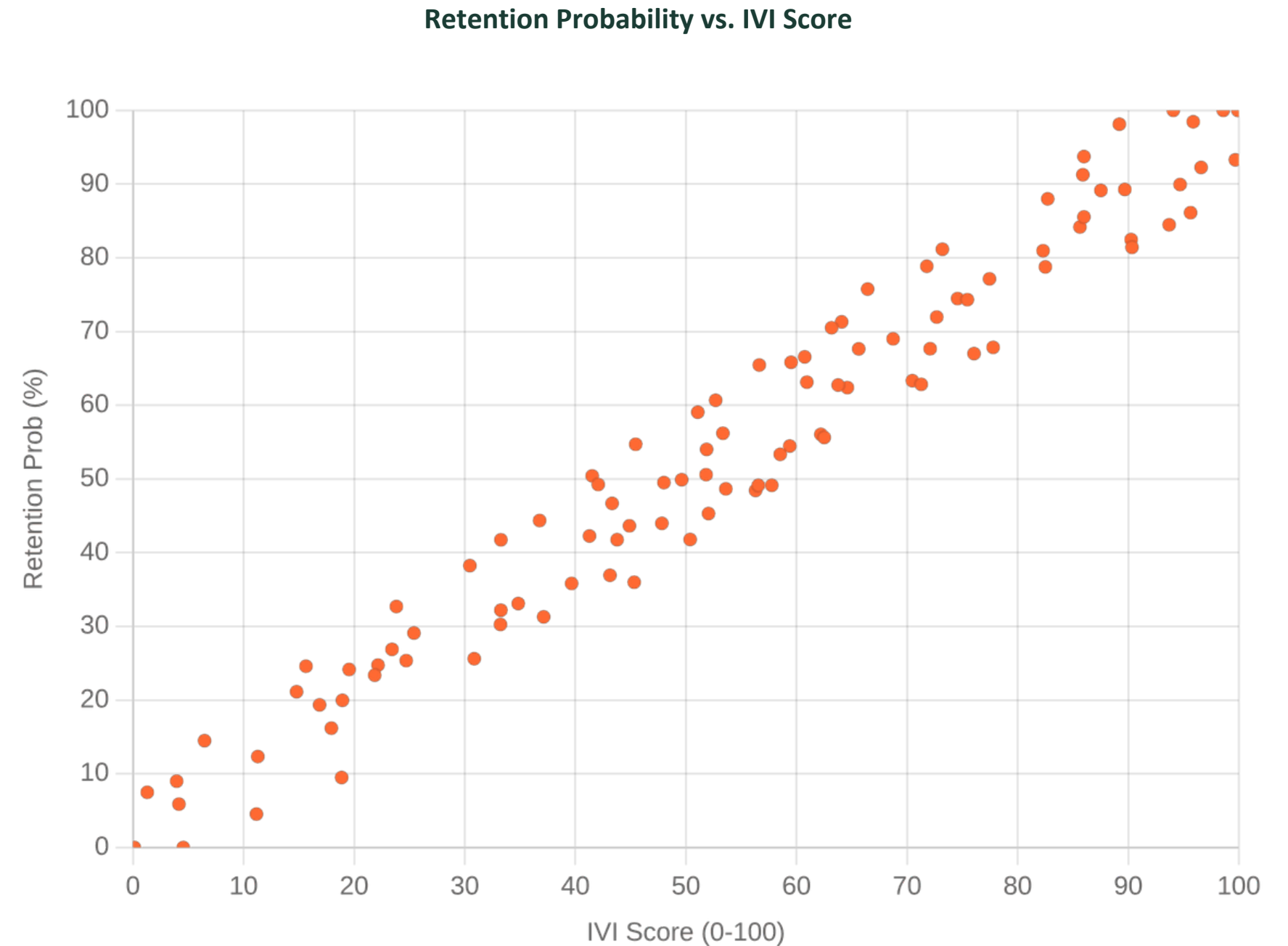
Probability of contract churn / non-renewal.

Model Approach

Predictive framework using IVI components and historical trends

Business Interpretation

Low IVI + Declining Retention Probability = High Churn Risk



Higher IVI scores (lower risk) strongly correlate with higher retention probability.

Retention probability is inferred using IVI level and experience trends as a proxy for churn risk.

Strategic Recommendations

Utilization Management	Provider Efficiency	Expected Impact
Implement cost-control programs targeting high-utilization contracts (59.1% weight).	Enhance network coordination for high-U segments to reduce services per episode.	Reduced medical cost leakage and improved contract retention rates.

Tactical Action Matrix

Risk Profile (IVI)	Retention Status	Recommended Action
High Risk (0-50)	Low Probability	Immediate account intervention & relationship management
High Risk (0-50)	High Probability	Cost optimization & utilization review
Low Risk (70-100)	High Probability	Upsell opportunities & contract expansion
Low Risk (70-100)	Low Probability	Experience improvement & engagement programs

Technical Documentation

Data Sources

Claims data (medical services, costs)
Member interaction (calls, complaints)
Contract & member demographics

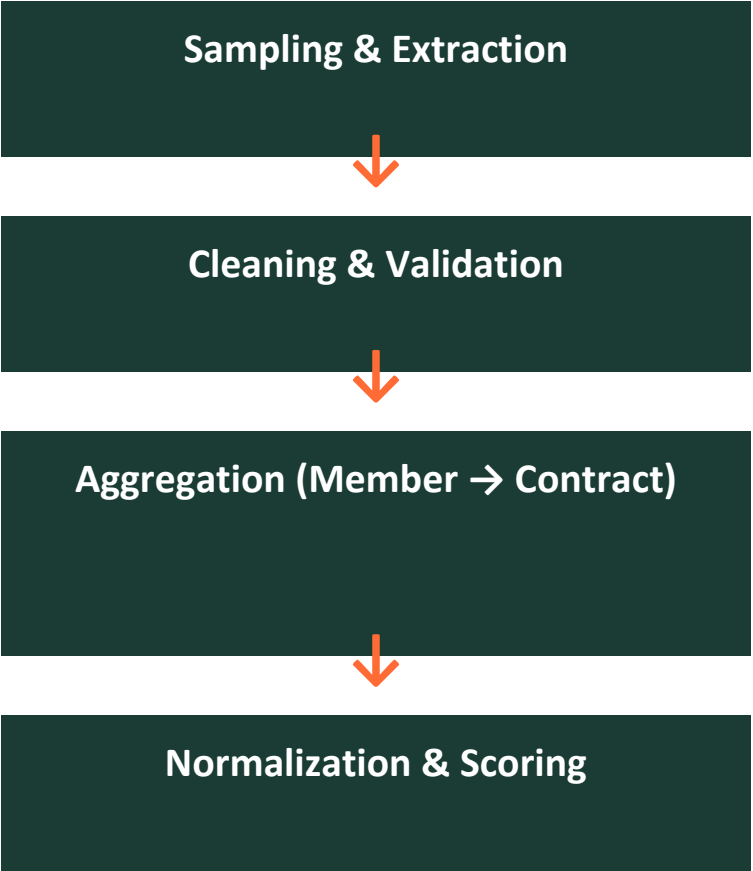
Key Assumptions

Z-score standardization for scoring
12-month rolling analysis period
Imputation using portfolio medians

Model Limitations

Historical bias in training data
Retraining required periodically

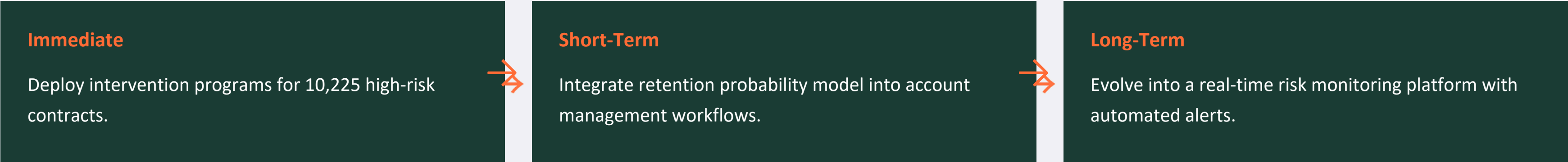
Data Processing Pipeline



Conclusion & Next Steps

The IVI framework successfully identifies that **utilization efficiency** drives portfolio sustainability. With 32.3% of contracts classified as high-risk, we now have a clear roadmap for intervention prioritization.

Strategic Roadmap



Transforming Reactive Risk into Proactive Intelligence.